

2021 Ph H2 Q5

Section: Our Dynamic Universe

Topic: Gravitation & Interference

Two neutron stars ($m_1 = 3.18 \times 10^{30}$ kg, $m_2 = 2.27 \times 10^{30}$ kg) merge, producing gravitational waves. (a) Calculate separation when gravitational force = 1.59×10^{39} N.

(b)(i) Explain formation of minimum in interferometer interference pattern. (b)(ii) Compare 4.0×10^{-18} m arm length change with original 4.0 km.

Worked solution

(a) Separation

$$\begin{aligned} F &= G m_1 m_2 / r^2 \Rightarrow r = \sqrt{(G m_1 m_2 / F)} \\ &= \sqrt{((6.67 \times 10^{-11})(3.18 \times 10^{30})(2.27 \times 10^{30}) / (1.59 \times 10^{39}))} \\ &= 5.50 \times 10^5 \text{ m} \end{aligned}$$

Answer: 1.1×10^6 m

(b)(i) Minimum in interference

When beams recombine, a minimum occurs if path difference = $(n + \frac{1}{2})\lambda$, so the waves arrive out of phase and undergo destructive interference.

(b)(ii) Order of magnitude comparison

$$\begin{aligned} \Delta L / L &= 4.0 \times 10^{-18} / 4.0 \times 10^3 = 1.00 \times 10^{-21} \\ &= 1 \times 10^{-21} \text{ (order of magnitude)}. \end{aligned}$$

Answer: change is about 10^{-21} times the original length

Final answers

(a) $r \approx 1.1 \times 10^6 \text{ m}$

(b)(i) Destructive interference (path difference = $(n + \frac{1}{2})\lambda$)

(b)(ii) $\Delta L/L \approx 10^{-21}$

Revision tips

- Newton's law of gravitation: $F = G m_1 m_2 / r^2$.
- Interferometer minima form by destructive interference.
- Path difference = $(n + \frac{1}{2})\lambda$ gives out-of-phase cancellation.
- Gravitational wave strains are incredibly small ($\sim 10^{-21}$).
- Comparisons of $\Delta L/L$ highlight extreme detector sensitivity.